Example 7.47

Prove that the function $f(x) = x^2 - 2x - 3$ is strictly increasing in $(2, \infty)$.

Solution

Since $f(x) = x^2 - 2x - 3$, $f'(x) = 2x - 2 > 0 \ \forall x \in (2, \infty)$. Hence $f(x)$ is strictly increasing in $(2, \infty)$.

7.6.2 Absolute maxima and minima

The absolute maxima and absolute minima are referred to describing the largest and smallest values of a function on an interval.

Definition 7.8

Let x_0 be a number in the domain D of a function $f(x)$. Then $f(x_0)$ is the absolute maximum value of $f(x)$ on D, if $f(x_0) \geq f(x) \forall x \in D$ and $f(x_0)$ is the absolute minimum value of $f(x)$ on D if $f(x_0) \leq f(x) \forall x \in D$.

In general, there is no guarantee that a function will actually have an absolute maximum or absolute minimum on a given interval. The following figures show that a continuous function may or may not have absolute maxima or minima on an infinite interval or on a finite open interval.

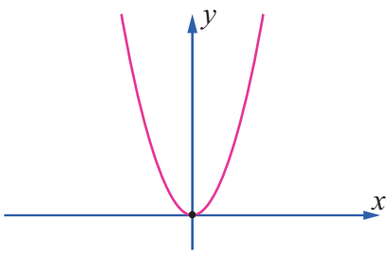


Fig. 7.15

$f(x)$ has an absolute minimum but no absolute maximum on $(-\infty, \infty)$

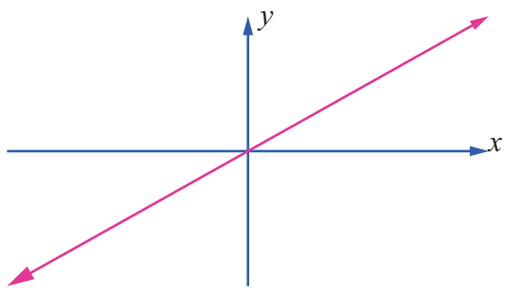


Fig. 7.16

$f(x)$ has no absolute extrema on $(-\infty, \infty)$.

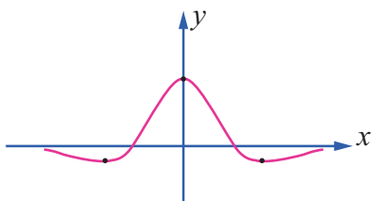


Fig. 7.17

$f(x)$ has an absolute maximum and an absolute minimum on $(-\infty, \infty)$

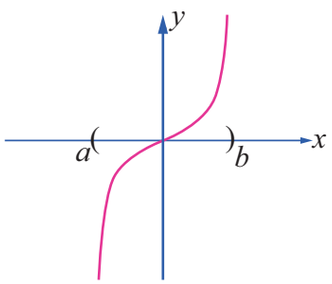


Fig. 7.18

$f(x)$ has no absolute extrema on (a, b).

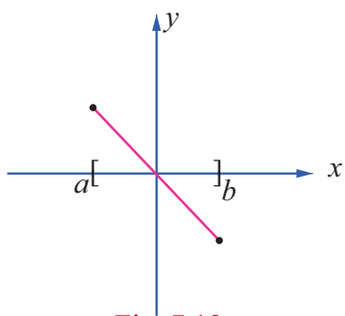


Fig. 7.19

$f(x)$ has an absolute maximum and an absolute minimum on $[a, b]$

However, the following theorem shows that a continuous function must have both an absolute maximum and an absolute minimum on every closed interval.

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